

Develop a logical data model reflecting the following requirements:

A. Derive relations (and FKs) from the conceptual model. Include the relational schema (e.g.: Staff(staffID, fName, lName, position, salary)).

- Outlet (outletNo, address, phone, fax)
- Vehicle (registrationNo, model, make, engineSize, capacity, mileage, dailyHireRate, outletNo {FK})
- Client (clientNo, fName, lName, address, phone, DOB, licenseNo)
- HireAgreement (hireNo, clientNo {FK}, name, address, phone, agreementStart, agreementEnd, registrationNo {FK}, model, make, startingMileage, endingMileage)
- Staff (staffNo, fName, lName, address, phone, DOB, sex, employmentStart, title, salary, outletNo {FK})

B. Validate the logical model using normalization to 3NF.

UNF => 1NF: Flatten the table. Without seeing the raw data, we cannot know whether it is unnormalized or in First Normal Form. For the purposes of this exercise, we will assume the data is in First Normal Form.

1NF => 2NF: Remove partial dependencies. The following are the functional dependencies:

- Outlet: outletNo => address, phone, fax (*full functional dependency*)
- Vehicle: registrationNo => model, make, engineSize, capacity, mileage, dailyHireRate, outlet (*full functional dependency*)
- Client: clientNo => fName, lName, address, phone, DOB, licenseNo (*full functional dependency*)
- Client: licenseNo => clientNo, fName, lName, address, phone, DOB (*full functional dependency*)
- HireAgreement: hireNo => clientNo, name, address, phone, agreementStart, agreementEnd, registrationNo, model, make, startingMileage
- HireAgreement: clientNo => name, address, phone (*transitive dependency*)
- HireAgreement: registrationNo => model, make, startingMileage (*transitive dependency*)
- Staff: staffNo => fName, lName, address, phone, DOB, sex, employmentStart, title, salary, outlet (*full functional dependency*)

Based on the primary keys identified, there are no partial dependencies, only transitive dependencies. Therefore, the data is already in Second Normal Form.

2NF => 3NF: Remove transitive dependencies. Restructure the HireAgreement table to remove the attributes contained in other tables that are linked via foreign keys:

HireAgreement (hireNo, clientNo {FK}, agreementStart, agreementEnd, registrationNo {FK}, startingMileage, endingMileage)

- C. Validate the logical model against 5 user transactions. (Note: These will be then implemented in 3c.)
- I. You must include the query (question) in plain English.
 1. List the client name (first and last), vehicle registration number, make and model of car, and outlet address and phone number for all hire agreements with starting date in February 2026.
 2. How many staff members are employed at the outlet that contains the first vehicle contracted in February 2026?
 3. What is the name (first and last) of the oldest staff member at the outlet that contracted the vehicle with the highest ending mileage in February 2026?
 4. What is the date of birth and the license number of the client that has rented vehicles from the greatest number of outlets since January 2020?
 5. Since January 2020, what are the average numbers of hire agreements per vehicle and hire agreements per client for outlets located in the state of Florida?
 - II. The majority of queries (questions) should involve more than one entity.
 - III. You must explain verbally what steps you follow using the relationships between the entities (no code!).
 1. List the client name (first and last), vehicle registration number, make and model of car, and outlet address and phone number for all hire agreements with starting date in February 2026.
 - AgreementStart is held in the Hire Agreement table, while fName and lName of clients are held in the Client table, registrationNo and make and model of car are held in the Vehicle table, and outlet address and phone are held in the Outlet table.
 - To produce the required list, use the Client *signs* Hire Agreement, Vehicle *included in* Hire Agreement, and Outlet *has* Vehicle relationships.
 - This transaction will leverage the following foreign keys to complete: clientNo in the Hire Agreement table for the Client *signs* Hire Agreement relationship, registrationNo in the HireAgreement table for the Vehicle *included in* Hire Agreement relationship, and

outletNo in the Vehicle table for the Outlet *has* Vehicle relationship.

2. How many staff members are employed at the outlet that contains the first vehicle contracted in February 2026?
 - StaffNo is held in the Staff table, outletNo is held in the Outlet table, and registrationNo and agreementStart are held in the Hire Agreement table.
 - To produce the required list, use the Outlet *employs* Staff, Outlet *has* Vehicle, and Vehicle *included in* Hire Agreement relationships.
 - This transaction will leverage the following foreign keys to complete: outletNo in the Staff table for the Outlet *employs* Staff relationship, outletNo in the Vehicle table for the Outlet *has* Vehicle relationship, and registrationNo in the Hire Agreement table for the Vehicle *included in* Hire Agreement relationship.
3. What is the name (first and last) of the oldest staff member at the outlet that contracted the vehicle with the highest ending mileage in February 2026?
 - Staff fName, lName, and DOB are held in the Staff table, outletNo is held in the Outlet table, registrationNo is held in the Vehicle table, and endingMileage and agreementStart are held in the Hire Agreement table.
 - To produce the required list, use the Outlet *employs* Staff, Outlet *has* Vehicle, and Vehicle *included in* Hire Agreement relationships.
 - This transaction will leverage the following foreign keys to complete: outletNo in the Staff table for the Outlet *employs* Staff relationship, outletNo in the Vehicle table for the Outlet *has* Vehicle relationship, and registrationNo in the Hire Agreement table for the Vehicle *included in* Hire Agreement relationship.
4. What is the date of birth and the license number of the client that has rented vehicles from the greatest number of outlets since January 2020?
 - Client DOB and licenseNo is held in the Client table, registrationNo is held in the Vehicle table, hireNo and agreementStart is held in the Hire Agreement table, and outletNo is held in the Outlet table.
 - To produce the required list, use the Client *signs* Hire Agreement, Outlet *has* Vehicle, and Vehicle *included in* Hire Agreement relationships.

- This transaction will leverage the following foreign keys to complete: clientNo in HireAgreement table for Client *signs* Hire Agreement relationship, outletNo in the Vehicle table for the Outlet *has* Vehicle relationship, and registrationNo in the Hire Agreement table for the Vehicle *included in* Hire Agreement relationship.
5. Since January 2020, what are the average numbers of hire agreements per vehicle and hire agreements per client for outlets located in the state of Florida?
- AgreementStart and hireNo are held in the Hire Agreement table, registrationNo is held in the Vehicle table, clientNo is held in the Client table, and outletNo and outlet address are held in the Outlet table.
 - To produce the required list, use the Client *signs* Hire Agreement, Vehicle *included in* Hire Agreement, and Outlet *has* Vehicle relationships.
 - This transaction will leverage the following foreign keys to complete: clientNo in the Hire Agreement table for the Client *signs* Hire Agreement relationship, registrationNo in Hire Agreement for the Vehicle *included in* Hire Agreement relationship, and outletNo in Vehicle table for the Outlet *has* Vehicle relationship.

D. Define integrity constraints:

I. Primary key constraints.

Primary keys must be unique and not null. The primary keys identified are outletNo for the Outlet table, registrationNo for the Vehicle table, clientNo for the Client table, hireNo for the HireAgreement table, and staffNo for the Staff table. These primary keys all allow you to uniquely identify tuples – the problem explicitly mentions this for registrationNo, hireNo, and clientNo, and we can assume it for outletNo because it is assigned by the company. Without seeing the raw data, we cannot know for certain that the primary keys do not have null values, but I will assume this.

II. Referential integrity/Foreign key constraints.

Referential integrity/foreign key constraints mean that if a FK contains a value, it must refer to existing tuple in its parent relation. In addition, if child relation participation is mandatory, then nulls are not allowed in the FK. There are also existence constraints that define under which conditions a FK may be inserted, updated, or deleted. In this case, the foreign keys are outletNo in the Vehicle

table, clientNo and registrationNo in the HireAgreement table, and outletNo in the Staff table. Therefore, outletNo in the Vehicle and Staff table must match an existing outletNo in the Outlet table, clientNo in the HireAgreement table must match an existing clientNo in the Client table, and registrationNo in the HireAgreement table must match an existing registrationNo in the Vehicle table. Regarding participation constraints, all relationships are 1:*, with mandatory participation on both sides:

- Outlet *has* Vehicle relationship: Vehicle is the child because it is the “many” side of the relationship, which means FK outletNo cannot have nulls.
- Outlet *employs* Staff relationship: Staff is the child because it is the “many” side of the relationship, which means FK outletNo cannot have nulls.
- Vehicle *included in* HireAgreement relationship: HireAgreement is the child because it is the “many” side of the relationship, which means FK registrationNo cannot have nulls.
- Client *signs* HireAgreement relationship: HireAgreement is the child because it is the “many” side of the relationship, which means FK clientNo cannot have nulls.

Below follows the existence constraints:

Outlet (outletNo, address, phone, fax)

Primary key: outletNo

Alternate key: address; alternate key: phone; alternate key: fax

Vehicle (registrationNo, model, make, engineSize, capacity, mileage, dailyHireRate, outletNo)

Primary key: registrationNo

Foreign key: outletNo **references** Outlet(outletNo) ON UPDATE CASCADE ON DELETE NO ACTION => *Cascade updates so that any changes in outlet number are captured for the vehicles in that outlet. Do not allow deletion of an outlet number until all its vehicles have been assigned to other outlets.*

Client (clientNo, fName, lName, address, phone, DOB, licenseNo)

Primary key: clientNo

Alternate key: licenseNo

Staff (staffNo, fName, lName, address, phone, DOB, sex, employmentStart, title, salary, outletNo)

Primary key: staffNo

Foreign key: outletNo **references** Outlet(outletNo) ON UPDATE CASCADE ON DELETE SET DEFAULT => *Cascade updates so that any changes in outlet number are captured for the staff*

that pertain to that outlet. When an outlet number is deleted (i.e., it closes), reassign the staff to the default (main) outlet.

HireAgreement (hireNo, clientNo, agreementStart, agreementEnd, registrationNo, startingMileage, endingMileage)

Primary key: hireNo

Alternate key: registrationNo, agreementStart

Foreign key: clientNo **references** Client(clientNo) ON UPDATE CASCADE ON DELETE NO ACTION => *Cascade updates so that any changes in client number are captured in associated hire agreements. Do not allow for the deletion of client numbers that have associated hire agreements.*

Foreign key: registrationNo **references** Vehicle(registrationNo) ON UPDATE CASCADE ON DELETE NO ACTION => *Cascade updates so that any changes in registration number are captured in the vehicle's hire agreements. Do not allow for the deletion of registration numbers that have associated hire agreements.*

III. Alternate key constraints (if any).

The alternate keys are addresss, phone, and fax for the Outlet relation (assuming no two outlets share an address or contact information and each outlet has a physical location), licenseNo for the Client relation, and composite key registrationNo, agreementStart for the HireAgreement relation. These should be unique and non-null.

IV. Required data.

Beyond the primary and alternate keys, attributes that should not hold nulls are all the attributes in the Vehicle table because clients will want to know the details of the cars that are available to hire (especially whether it is in the outlet where they are looking). All the attributes of the Outlet table should also be filled, perhaps except for fax IF you remove this as an alternate key. All the Client information should also be non-null – address so you know where to bill, phone so you can contact the client should there be any issue, DOB so you know client is old enough to rent the car, licenseNo so you know the client can legally drive, and name so you know who is signing the agreement. In the HireAgreement, all attributes are necessary for billing and tracking the cars. In the Staff table, perhaps sex, address (if tax, health insurance, and other documents are paperless), phone, employmentStart, or title could allow nulls, but realistically an employer would want a value for all of these.

V. Attribute domain constraints.

Table	Attributes	Data Type & Length	Nulls
Outlet	outletNo	Integer 1-99999	No
	address	255 variable characters	No

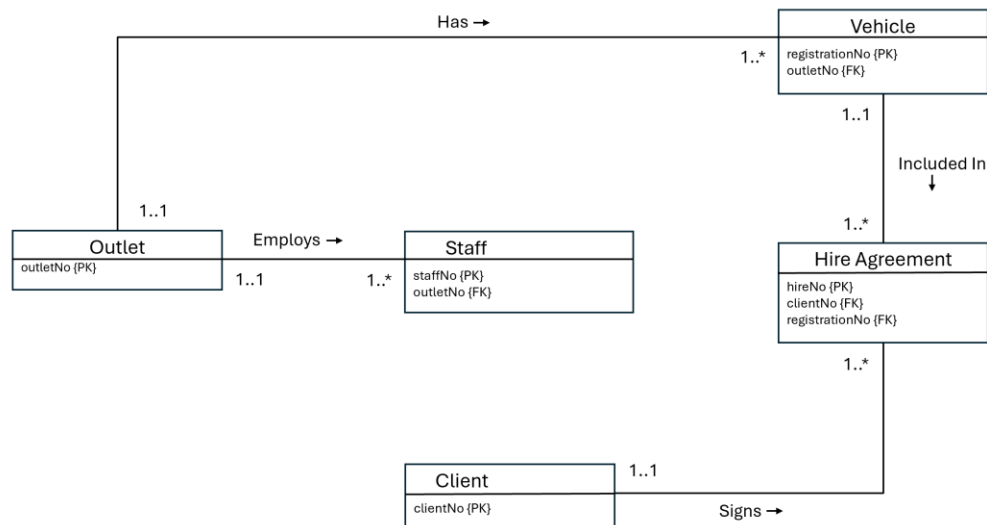
	phone	20 variable characters, in valid U.S./international phone pattern	No
	fax	20 variable characters, in valid U.S./international fax pattern	No
Vehicle	registrationNo	15 variable characters, alphanumeric	No
	model	50 variable characters, alphanumeric + spaces/hyphens	No
	make	30 variable characters, alphanumeric + spaces	No
	engineSize	Decimal .1-10.0	No
	capacity	Integer 1-20	No
	mileage	Integer 0-999999	No
	dailyHireRate	Decimal .01-9999.99	No
Client	outletNo {FK}	Integer 1-99999	No
	clientNo	Integer 1-999999	No
	fName	15 variable characters	No
	lName	15 variable characters	No
	address	255 variable characters	No
	phone	20 variable characters, in valid U.S./international phone pattern	No
	DOB	Date, must be < current date	No
HireAgreement	licenseNo	25 variable characters, alphanumeric	No
	hireNo	Integer 1-999999	No
	clientNo {FK}	Integer 1-999999	No
	agreementStart	Date, must be <= current date	No
	agreementEnd	Date, must be > current date	No
	registrationNo {FK}	15 variable characters, alphanumeric	No
	startingMileage	Integer 0-999999	No
Staff	endingMileage	Integer 0-999999	No
	staffNo	Integer 1-999999	No
	fName	15 variable characters	No
	lName	15 variable characters	No
	address	255 variable characters	No
	phone	20 variable characters, in valid U.S./international phone pattern	No
	DOB	Date, must be < current date	Yes
	sex	1 character (M or F)	Yes

	employmentStart	Date, must be <= current date	No
	title	50 variable characters	No
	salary	Decimal 10000.00–500000.00	No
	outletNo {FK}	Integer 1-99999	No

VI. General constraints (if any). (Constraints governing “real-world” transactions; business-applied constraints)

The problem does not specify any general constraints. However, I imagine that were this real life, a general constraint could be that each outlet has the capacity for only a certain number of vehicles, so that vehicles can be reassigned to other outlets if capacity is almost reached. In addition, certain clients could be blacklisted from taking on hire agreements and their clientNos unallowable values on the agreements. Other general constraints could be that the Hire Agreement can only span a certain number of days, that certain job titles correspond to certain salary ranges, or that clients must be over a certain age (usually 21).

E. Generate the E-R diagram for the logical level (contains FKs as attributes).



Notes: A report must be created including detailed documentation of each of the steps addressing all the required items. ER diagrams for the logical model must be included in the report. All assumptions made in the design must be clearly stated.